

# Honors Calculus 1610

## Lecture 1: Mathematical Context, Sets, and Functions

Fall 2026

# What is this course about?

## Our main goal

To understand calculus in the context of modern, rigorous mathematical thought: not only how to use calculus, but how it is built and how we know that it works.

- ▶ We will use multivariable calculus as a setting in which to practice.
- ▶ We will read, evaluate, and construct mathematical arguments.
- ▶ A typical class alternates among lecture, collaborative worksheet work, and further lecture.

# A brief historical perspective

**ca. 300 BCE**

Euclid's *Elements*  
axiomatic plane geometry

**1700–1900**

major new developments  
in mathematics

**ca. 1900**

set theory becomes  
a common foundation

- ▶ Euclid's geometry gave a powerful axiomatic account of plane geometry and a geometric way to think about number.
- ▶ Complex numbers, non-Euclidean geometry, infinite series, and calculus strained that older foundation.
- ▶ Cantor, Dedekind, and Hilbert helped make set theory a common language for modern mathematics.

# Why set theory?

Since around 1900, set theory has been the basic foundational structure of mathematics. It provides a common language for discussing numbers, functions, and the objects on which calculus acts.

## Working definition

A **set** is an unordered collection of objects, called its **elements**. A set is determined by the elements it contains.

Therefore, order and repetition do not matter:

$$\{1, 2, 3\} = \{3, 2, 1\}, \quad \{1, 1, 2\} = \{1, 2\}.$$

# Membership and standard sets

## Membership notation

$a \in A$  means that  $a$  is an element of  $A$ ,       $a \notin A$  means that it is not.

$$1 \in \{1, 2, 3\}, \quad 4 \notin \{1, 2, 3\}.$$

## Standard sets

$$\mathbb{N} = \{0, 1, 2, 3, \dots\}, \quad \mathbb{R} = \text{the set of real numbers.}$$

# Set-builder notation

The notation

$$\{n \in \mathbb{N} \mid n \text{ is even}\}$$

means the set of natural numbers  $n$  such that  $n$  is even. The vertical bar is read “such that.”

For example,

$$\{n \in \mathbb{N} \mid n \text{ is odd and } 1 \leq n \leq 9\} = \{1, 3, 5, 7, 9\}.$$

# Operations on sets

If  $A$  and  $B$  are sets, then

$$A \cup B = \{x \mid x \in A \text{ or } x \in B\},$$

(union),

$$A \cap B = \{x \mid x \in A \text{ and } x \in B\},$$

(intersection),

$$A \setminus B = A - B = \{x \mid x \in A \text{ and } x \notin B\},$$

(difference).

The set  $A \setminus B$  makes sense whether or not  $B \subseteq A$ .

# Subsets and equality

Subset

$A \subseteq B \iff$  every element of  $A$  is an element of  $B$ .

Equivalently,

$$x \in A \implies x \in B.$$

The fundamental criterion for equality of sets

$$A = B \iff A \subseteq B \text{ and } B \subseteq A.$$

To prove that two sets are equal, prove both containments.



# Functions

If  $X$  and  $Y$  are sets, a function

$$f: X \longrightarrow Y$$

is a rule which associates to every  $x \in X$  a uniquely determined element  $f(x) \in Y$ .

Images and inverse images

If  $S \subseteq X$ , then

$$f(S) = \{f(s) \mid s \in S\}.$$

If  $T \subseteq Y$ , then

$$f^{-1}(T) = \{x \in X \mid f(x) \in T\}.$$

For  $y \in Y$ , we write

$$f^{-1}(y) = f^{-1}(\{y\}) = \{x \in X \mid f(x) = y\}.$$

Example:  $f(x) = x^2$

Let  $f: \mathbb{R} \rightarrow \mathbb{R}$  be given by  $f(x) = x^2$ .

$$f([0, 1]) = [0, 1],$$

$$f^{-1}([0, 1]) = [-1, 1],$$

$$f^{-1}(4) = f^{-1}(\{4\}) = \{-2, 2\}.$$

# Proposition

Let  $f: X \rightarrow Y$ , and let  $A, B \subseteq Y$ . Then

$$f^{-1}(A \cup B) = f^{-1}(A) \cup f^{-1}(B).$$

To prove equality, prove both inclusions:

$$\begin{aligned} f^{-1}(A \cup B) &\subseteq f^{-1}(A) \cup f^{-1}(B), \\ f^{-1}(A) \cup f^{-1}(B) &\subseteq f^{-1}(A \cup B). \end{aligned}$$

## Proof: first containment

Suppose that  $x \in f^{-1}(A \cup B)$ . Then

$$f(x) \in A \cup B.$$

Therefore,

$$f(x) \in A \quad \text{or} \quad f(x) \in B.$$

Thus,

$$x \in f^{-1}(A) \quad \text{or} \quad x \in f^{-1}(B),$$

which says

$$x \in f^{-1}(A) \cup f^{-1}(B).$$

Hence

$$f^{-1}(A \cup B) \subseteq f^{-1}(A) \cup f^{-1}(B).$$

## Proof: second containment

Suppose that  $x \in f^{-1}(A) \cup f^{-1}(B)$ . Then

$$x \in f^{-1}(A) \quad \text{or} \quad x \in f^{-1}(B).$$

Therefore,

$$f(x) \in A \quad \text{or} \quad f(x) \in B.$$

Thus,

$$f(x) \in A \cup B,$$

so

$$x \in f^{-1}(A \cup B).$$

Hence

$$f^{-1}(A) \cup f^{-1}(B) \subseteq f^{-1}(A \cup B).$$

## Conclusion

Since both containments hold,

$$f^{-1}(A \cup B) = f^{-1}(A) \cup f^{-1}(B).$$